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(54) **FLEXIBLE ABRASIVE PRODUCT**
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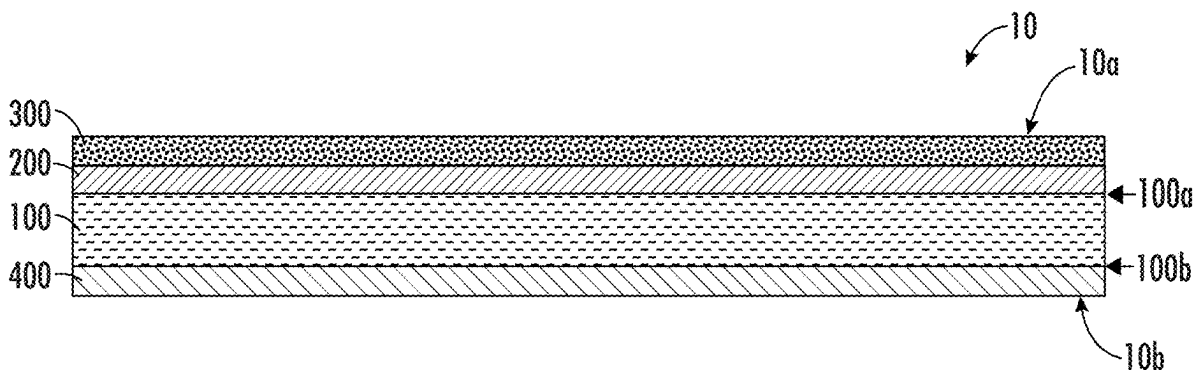
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(57) **ABSTRACT**

A flexible abrasive product includes a woven textile substrate having a first surface and a second surface, where the woven textile substrate contains a plurality of continuous multi-filament yarns. The flexible abrasive product also includes a first adhesive layer on the first surface of the woven textile substrate and a plurality of abrasive particles both on and at least partially in the first adhesive layer.



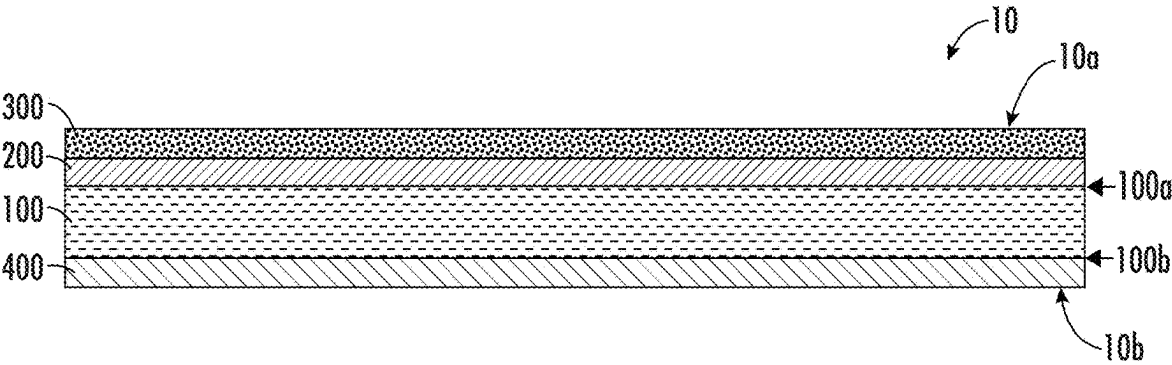


FIG. 1

FLEXIBLE ABRASIVE PRODUCT

RELATED APPLICATIONS

[0001] This application claims priority to co-pending U.S. Provisional Patent Application 63/562,404 filed on Mar. 7, 2024, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD OF THE INVENTION

[0002] This patent application relates to flexible abrasive products, such as those in sanding belts.

BACKGROUND

[0003] Many abrasive products have a textile within or on the back of the composite such as in a sanding disk, belt, or other sanding product. In many cases, the textile is made solely of spun yarns in a woven or scrim construction. It would be beneficial to have a some additional properties of the fabric such as thinness, strength, flatness, and/or low cost by utilizing multifilament yarns in the textile.

BRIEF SUMMARY OF THE INVENTION

[0004] In some aspects, the techniques described herein relate to a flexible abrasive product including: a woven textile substrate having a first surface and a second surface, wherein the woven textile substrate includes a plurality of continuous multi-filament yarns, a first adhesive layer on the first surface of the woven textile substrate; and a plurality of abrasive particles both on and at least partially in the first adhesive layer.

BRIEF DESCRIPTION OF THE FIGURES

[0005] FIG. 1 shows an illustrative cross-section of one embodiment of a flexible abrasive product.

DETAILED DESCRIPTION OF THE INVENTION

[0006] Referring to FIG. 1, there is shown one embodiment of the flexible abrasive product 10. This flexible abrasive product 10 has a first side 10a and a second side 10b and contains a plurality of layers. The flexible abrasive product may be used for any suitable end use and is most often used in sanding applications. The flexible abrasive product 10 could be, for example, an endless abrasive belt, a disc, or a rotatable drum. Most applications need grit to sand something down and some structural integrity given by a woven textile substrate.

[0007] The flexible abrasive product 10 contains a woven textile substrate 100 having a first surface 100a and a second surface 100b. The woven textile substrate 100 is referred to as woven, but in other embodiments could be a knit or nonwoven textile.

[0008] Preferably, the woven textile substrate 10 is a woven textile and may be in any suitable weave pattern. A twill weave pattern has been shown to be preferred for some abrasion product applications. In some embodiments, a twill weave is preferred because of flexibility and flatness of each surface of the textile. If more flexibility and/or flatness of each surface is desired, a sateen weave can also be used in some applications.

[0009] The twill weave preferably has between 50 and 120 ends per inch and 25 and 80 picks per inch. Why is this range

preferred? Typically, abrasive substrates have twice as many ends vs picks due to the need for a higher degree of tensile strength that is needed in the machine direction vs the cross-machine direction. Overall, the woven textile substrate has an areal weight range of 4-16 osy (ounces per square yard). It has been found that this range produces a textile that has strength and durability while minimizing the weight.

[0010] Preferably, the woven textile substrate 100 contains continuous multi-filament yarns. It has been found that these yarns produce a textile that is flatter and smoother than other textiles made with staple yarns. Preferably, less than about 5% of the fibers within the yarns of the woven textile substrate are staple yarns having a length of less than about 3 inches. It is preferred to not intentionally include any staple fibers, but a small percentage of the continuous multi-filament fibers may break during formation of the fibers, yarns, textile, and/or product.

[0011] In one embodiment, the plurality of continuous multi-filament yarns contains polyester terephthalate (PET). PET yarns are preferred as they are low cost and high strength. In one embodiment, the plurality of continuous multi-filament yarns consists essentially of PET. "Consists essentially of" in this application is defined to mean at least 95% by weight.

[0012] In one embodiment, the polyester terephthalate multi-filament yarns are texturized. Textured yarn has been processed such that the individual filaments have been crimped and/or air-entangled to each other. This causes the yarn to mimic the feel of staple spun yarn. In this context, it is preferred because of the typically increased tensile strength per linear density. Preferably the polyester terephthalate multi-filament yarns are low elongation (filament yarns that have been drawn more vs industry standard). Having this low elongation is preferred to lessen or eliminate the stretch or deformation the flexible abrasion product 10 during use. Preferably, the woven textile substrate has an elongation at a load of 44 lbs of less than about 4%. Also preferably, the flexible abrasive product has an elongation at a load of 100 lbs of less than about 8%. This is measured on an Instron machine and reading data from the stress/strain curve. It is important for the life of the abrasive belt and the way it fits.

[0013] While in some embodiments it is preferred that the yarns making up the woven textile substrate 100 are PET, they may also be any other suitable material. The yarns in the textile may also comprise additional fibers including, but not limited to polyester fibers (e.g., poly(ethylene terephthalate) fibers, poly(propylene terephthalate) fibers, poly(trimethylene terephthalate) fibers), poly(butylene terephthalate) fibers, and blends thereof), polyamide fibers (e.g., nylon 6 fibers, nylon 6,6 fibers, nylon 4,6 fibers, and nylon 12 fibers), polyvinyl alcohol fibers, and combinations, mixtures, or blends thereof. The yarn(s) can include other synthetic fibers, such as static dissipative or antistatic fibers. In one embodiment, the yarns in the woven textile substrate may contain staple yarns.

[0014] In some embodiment, the yarns can also comprise natural fibers, such as cotton, linen, jute, hemp, or wool. The yarns can also comprise other fibers, such as rayon, lyocell, or acetate. When such fibers (e.g., cotton fibers) are present in the woven textile substrate 100 of the invention, it may be desirable to treat the woven textile substrate 100 with a flame retardant in order to impart some degree of flame

resistance to these fibers and produce a flame-resistant textile exhibiting a desired degree of flame resistance.

[0015] The woven textile substrate **100** has yarns running in two, generally perpendicular directions typically called a warp or machine direction and weft or fill or cross-machine direction. All of the yarns within the woven textile substrate **100** may contain the same materials or they may be different. For example, the woven textile substrate may have continuous multi-filament PET yarns in the warp direction and continuous or staple nylon yarns in the weft direction. Even within the same direction, the woven textile substrate may have multiple types of yarns preferably in a pattern (such as 3 PET yarns and then 1 nylon/cotton yarn).

[0016] The woven textile substrate **100** of the invention can have any suitable weight (i.e., weight per unit area). The woven textile substrate **100** preferably has a weight of about 16 oz/yd² or less (about 540 g/m² or less), about 14 oz/yd² or less (about 470 g/m² or less), about 12 oz/yd² or less (about 410 g/m² or less), about 10 oz/yd² or less (about 340 g/m² or less), about 9 oz/yd² or less (about 310 g/m² or less). More preferably, the woven textile substrate **100** has a weight of about 8 oz/yd² or less (about 270 g/m² or less), more preferably about 7 oz/yd² or less (about 240 g/m² or less), more preferably about 6.5 oz/yd² or less (about 220 g/m² or less), more preferably about 6 oz/yd² or less (about 200 g/m² or less), more preferably about 5.75 oz/yd² or less (about 195 g/m² or less), and most preferably about 5.5 oz/yd² or less (about 190 g/m² or less). In one embodiment, the woven textile substrate **100** preferably has a weight of between about 7 oz/yd² (about 240 g/m²) to about 16 oz/yd² (about 540 g/m²).

[0017] On the first surface **100a** of the woven textile substrate **100** is a first adhesive layer **200**. The purpose of the first adhesive layer **200** is to attach the abrasive particles **300** to the woven textile substrate **100**. The adhesive is sometimes referred to as a binder or glue.

[0018] The adhesive in the first adhesive layer may be any suitable adhesive that serves to attach the textile to the particles and function well in the end product. In one embodiment, the first adhesive layer preferably contains a thermoset polymer or a thermoplastic polymer (or a mixture of both). In one embodiment, the first adhesive layer contains a polymer selected from the group consisting of phenolic resins, epoxy resins, polyester resins, copolyester resins, polyurethane resins, polyamide resins and mixtures thereof.

[0019] The first adhesive layer **200** may be applied to the woven textile substrate **100** in any conventionally known coating system such as gravure coating, reverse roll coating, die coating, air doctor coating, blade coating, rod coating, bar coating, curtain coating, knife coating, transfer roll coating, squeeze coating, dip coating, kiss coating, spray coating, calender coating, and extrusion coating.

[0020] In one embodiment, the flexible abrasive product **10** contains a second adhesive layer **400** on the second surface **100b** of the woven textile substrate **100**. This second adhesive layer **400** may serve to form the backside of the flexible abrasive product **10** and may provide a smoother surface, surface with less yarns or fibers by coating the textile, a printable surface, or any other benefit. Preferably, the materials selected for the second adhesive layer **400** are from the same listing as for the first adhesive layer **200**. The second adhesive layer **400** may be the same thickness or thinner, or thicker than the first adhesive layer **200**. The

flexible abrasive product **10** may also contain additional layers (such as additional textile layers, film layers, or adhesive layers) attached to the second adhesive layer **400** for various desired properties.

[0021] In one embodiment, the flexible abrasive product **10** contains a primer layer located between the woven textile substrate **100** and the first adhesive layer **200**. This primer layer may help with adhesion between the two layers and may also help control the amount of adhesive that soaks into the woven textile substrate.

[0022] The flexible abrasion product **10** also contains a plurality of abrasive particles **300** both on and at least partially in the first adhesive layer **200**. These abrasive particles **300** are the grit on sandpaper like products. The plurality of abrasive particles **300** wherein the plurality of abrasive particles has an average abrasive particle size in the range of about 2 μ m to about 750 μ m.

[0023] The preferred abrasive particles are preferably selected from the group consisting of fused aluminum oxide, ceramic aluminum oxide, sol gel alumina-based ceramics, silicon carbide, glass, ceria, glass ceramics, fused alumina-zirconia, natural crushed aluminum oxide, heat treated aluminum oxide, zirconia, garnet, emery, cubic boron nitride, diamond, particulate polymeric materials, metals and combinations and agglomerates thereof. The abrasive particle may also have a shape associated with it. Examples of such shapes include rods, triangles, pyramids, cones, solid spheres, hollow spheres and the like. Alternatively, the abrasive particle may be randomly shaped. The abrasive particles can be coated with materials to provide the particles with desired characteristics. For example, materials applied to the surface of an abrasive particle have been shown to improve the adhesion between the abrasive particle and the polymer. At least a portion of the abrasion particles **300** are preferably at least partially embedded into the first adhesive layer **200** to anchor the particles and help prevent them from falling of the flexible abrasion product prematurely. In one embodiment, the plurality of particles also contains an additional adhesive to hold the particles together and on the product.

[0024] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0025] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the subject matter of this application (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the subject matter of the

application and does not pose a limitation on the scope of the subject matter unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the subject matter described herein.

[0026] Preferred embodiments of the subject matter of this application are described herein, including the best mode known to the inventors for carrying out the claimed subject matter. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the subject matter described herein to be practiced otherwise than as specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the present disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

1. A flexible abrasive product comprising:
 - a woven textile substrate having a first surface and a second surface, wherein the woven textile substrate comprises a plurality of continuous multi-filament yarns,
 - a first adhesive layer on the first surface of the woven textile substrate; and
 - a plurality of abrasive particles both on and at least partially in the first adhesive layer.
2. The flexible abrasive product of claim 1, wherein the plurality of continuous multi-filament yarns comprises polyester terephthalate.
3. The flexible abrasive product of claim 1, wherein the plurality of continuous multi-filament yarns consists essentially of polyester terephthalate.
4. The flexible abrasive product of claim 1, wherein the plurality of continuous multi-filament yarns are texturized.

5. The flexible abrasive product of claim 1, wherein the woven textile substrate is in a twill weave.

6. The flexible abrasive product of claim 1, wherein the plurality of continuous multi-filament yarns of the woven textile substrate comprise less than about 5% staple length fibers having a length of less than about 3 inches.

7. The flexible abrasive product of claim 1, further comprising a second adhesive layer on the second surface of the woven textile substrate.

8. The flexible abrasive product of claim 1, wherein the first adhesive layer comprises a thermoset polymer or a thermoplastic polymer.

9. The flexible abrasive product of claim 1, wherein the first adhesive layer comprises a polymer selected from the group consisting of phenolic resins, epoxy resins, polyester resins, copolyester resins, polyurethane resins, polyamide resins and mixtures thereof.

10. The flexible abrasive product of claim 1, wherein the plurality of abrasive particles has an average abrasive particle size of about 2 μm to about 750 μm .

11. The flexible abrasive product of claim 1, wherein the flexible abrasive product is an endless abrasive belt.

12. The flexible abrasive product of claim 1, wherein the flexible abrasive product is a disc.

13. The flexible abrasive product of claim 1, wherein the flexible abrasive product is a rotatable drum.

14. The flexible abrasive product of claim 1, wherein the flexible abrasive product further comprises a primer layer on the woven textile substrate between the woven textile substrate and the first adhesive layer.

15. The flexible abrasive product of claim 1, wherein the woven textile substrate has an elongation at a load of 44 lbs of less than about 4%.

16. The flexible abrasive product of claim 1, wherein the flexible abrasive product has an elongation at a load of 100 lbs of less than about 8%.

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